# MATH 243 — Probability and Statistics

## Measures of Central Tendency: Mean, Median, Quartiles, Mode

*Lesson notes · Exam approach · TF practice with solutions · Quick notes (Statistics portion)*

## How to Read These Notes

| | |
|---|---|
| **EXAM FOCUS** | Exam importance, where it appears in the TF collection |
| **MEMORIZE** | Rules, tables and formulas to memorize |
| **TRAP** | Common mistakes and exam traps |
| **EXAM-READY ANSWER** | Model wording you can write in the exam |
| **NOTE** | Extra explanation / clarification |
| **Grey box, monospace** | Code, worked tables, derivations and hand-drawn sketches |
| **Centered grey box** | Formula to remember / final numerical result |
| **Purple heading / ✓ green** | Practice questions / their solutions |

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**Lesson 7 — Mode, Empirical Relation and Missing Frequencies** — Δ formula, Mode = 3Md − 2Mean, two unknown frequencies

**Quick Notes — Central Tendency** — revision bullets

> **NOTE** Page numbers refer to 2. Central Tendency.pdf (26 pp.), written as p. n. "TF p." = merged question PDF page. Mean, median and mode of grouped data are also needed inside every skewness question (Skewness, Moments and Kurtosis note).

# Exam Map — What the TF Papers Ask

| Session (paper) | Questions answered by this note [lesson] |
|---|---|
| 2021-22 (MATH 245) TF p. 44 | Q5(a) Monthly wages 1900-2400 (7 classes): suitable diagram, median from graph and formula; workers earning 2000–2170 (18) [L6] |
| 2015-16 (MATH 241) TF p. 15 | Q6(a) Incomplete distribution, N = 1000, median 415: missing frequencies, modal value, shape (18) [L7] |
| 2014-15 (MATH 241) TF p. 10 | Q6(a) Incomplete distribution, N = 1000, median 413.11: find the two missing frequencies and the mode (17) [L7] |

**EXAM FOCUS** 3 direct questions (53 marks), but grouped mean/median/mode appear inside every skewness and CV question too, so this deck supports about a third of the Statistics marks. The "missing frequencies from the median" problem was asked in 2014-15 and 2015-16 with the same table; the ogive-median problem in 2021-22. Know the median and mode formulas by heart.

# Lesson 5 — Arithmetic, Geometric and Harmonic Means

*pp. 1–6, 22*

A measure of central tendency is a single value around which the data tend to cluster; it represents the whole data and makes comparison easy. Types: arithmetic mean, median, mode, geometric mean, harmonic mean (p. 1).

$$\textbf{Ungrouped: } \bar{x} = \Sigma x_i/N; \quad \textbf{frequency distribution: } \bar{x} = \Sigma f_i x_i/N$$
$$\textbf{Short-cut: } \bar{x} = A + (\Sigma f_i d_i/N) \times h, \; d_i = (x_i - A)/h$$

A = assumed mean (the middle mid-value), h = class width, $x_i$ = mid-values, N = Σf.

| Marks | Mid x | f | d = (x − A)/h | fd |
|---|---|---|---|---|
| 0–20 | 10 | 4 | −3 | −12 |
| 20–40 | 30 | 26 | −2 | −52 |
| 40–60 | 50 | 22 | −1 | −22 |
| 60–80 | 70 | 10 | 0 | 0 |
| 80–100 | 90 | 9 | 1 | 9 |
| 100–120 | 110 | 6 | 2 | 12 |
| 120–140 | 130 | 3 | 3 | 9 |
| Total | | N = 80 | | Σ = −56 |

Marks of 80 students (p. 4): $\bar{x} = 70 + (-56/80) \times 20 = 56$. Other slide examples: incomes of 10 employees Σ = 44,940 → Tk 4494; discrete table → 320/25 = 12.8; 0–50 table → 25 + (−2/50)10 = 24.6; heights (boundaries 129.5–164.5) → 147 + (−33/100)5 = 145.35; practice answers 69.18 and 51.75.

$$\textbf{G.M.} = \text{antilog}(\Sigma f \log x / N); \quad \textbf{H.M.} = N / \Sigma(f/x)$$

**TRAP** With limits like 130–134, 135–139 use boundaries 129.5–134.5 … for the median/mode L and h, but the mid-values (132, 137, …) are the same either way.

# Lesson 6 — Median, Quartiles, Deciles, Percentiles and the Ogive

*pp. 7–18*

## 1. Median (pp. 7–13)

Ungrouped: arrange in order; N odd → ((N + 1)/2)-th value; N even → mean of the (N/2)-th and (N/2 + 1)-th values (weights 39 … 61 → 44; ten mothers → (43 + 44)/2 = 43.5).

$$\text{Median} = L + [(N/2 - \text{p.c.f.})/f] \times h$$

L = lower boundary of the median class (where the (N/2)-th value lies), p.c.f. = cumulative frequency before it, f = its frequency, h = width.

**Example:** Calculate the median for the distribution of the weights of 150 students from the given below:

| Weight | 30-40 | 40-50 | 50-60 | 60-70 | 70-80 | 80-90 |
|---|---|---|---|---|---|---|
| Frequency | 18 | 37 | 45 | 27 | 15 | 8 |

**Solution:**

| Weight | Frequency | Cumulative frequency |
|---|---|---|
| 30-40 | 18 | 18 |
| 40-50 | 37 | 55 → p.c.f |
| L ← 50-60 | 45 → f | 100 } 55 – 100 observation |
| 60-70 | 27 | 127 |
| 70-80 | 15 | 142 |
| 80-90 | 8 | 150 |
| | N = 150 | |

Median is $\frac{N}{2} = \frac{150}{2} = 75$ th observation. 75 th observation lies in class 50 – 60.

Median class is 50 – 60.

$$\text{Median} = L + \frac{\frac{N}{2} - \text{p.c.f}}{f} \times h = 50 + \frac{\frac{150}{2} - 55}{45} \times 10 = 54.44$$

*Figure 6.1 — Locating the median class (75th value) and applying the formula (p. 9)*

Slide answers: weights 54.44; overtime 20 + (50 − 31)/35 × 5 = 22.71 hrs (50 % work up to 22.71 hrs); profits 30 + (62.5 − 40)/36 × 10 = 36.25 crore; heights 144.5 + (50 − 48)/24 × 5 = 144.92; days absent 10 + (327.5 − 224)/241 × 5 = 12.15; practice 49.09 and 47.58.

> **NOTE** The profit example (p. 11) prints 36.5; the formula gives 30 + 22.5/36 × 10 = 36.25 crore.

## 2. Quartiles, Deciles, Percentiles (pp. 15–18)

$$Q_i = L + [(iN/4 - \text{p.c.f.})/f]\ h,\quad D_i: iN/10,\quad P_i: iN/100$$

Profits of 100 companies: $Q_1 = 40 + (25 - 12)/18 \times 10 = 47.22$; $Q_2 = 50 + (50 - 30)/30 \times 10 = 56.67$; $Q_3 = 60 + (75 - 60)/15 \times 10 = 70$ lakhs (25 % earn 47.22 lakhs or less). Overtime $Q_1 = 18.5$.

## 3. Graphical Median from the Ogive (textbook; asked 2021-22 Q5(a))

Draw the less-than ogive (upper boundary, cumulative frequency). Mark N/2 on the y-axis, go horizontally to the curve and drop to the x-axis: that x is the median. (Drawing both less-than and more-than ogives, they cross at the median.)

## Exam Approach

1. Cumulative frequency column; position N/2 (or iN/4); locate the class.
2. Substitute L, p.c.f., f, h in the formula; one line of arithmetic.
3. Interpret: "50 % of … are below …".

## Practice — Lesson 6

*(a) Actual — 2021-22 Q5(a) (TF p. 44).*

**(a)** Monthly wages of workers: 1900–1950: 6, 1950–2000: 10, 2000–2050: 22, 2050–2100: 30, 2100–2150: 16, 2150–2200: 12, 2200–2400: 15. Draw a suitable diagram, read the median from the graph and verify it by formula. How many workers earned monthly wages between 2000 and 2170 Tk? (18)

### ✓ Solution

| Wages | f | Less than | c.f. |
|---|---|---|---|
| 1900–1950 | 6 | 1950 | 6 |
| 1950–2000 | 10 | 2000 | 16 |
| 2000–2050 | 22 | 2050 | 38 |
| 2050–2100 | 30 | 2100 | 68 |
| 2100–2150 | 16 | 2150 | 84 |
| 2150–2200 | 12 | 2200 | 96 |
| 2200–2400 | 15 | 2400 | 111 |

Suitable diagram: the less-than ogive — plot (1900, 0), (1950, 6), (2000, 16), (2050, 38), (2100, 68), (2150, 84), (2200, 96), (2400, 111) and join. (The last class is wider, which an ogive handles without adjustment; a histogram would need height 15/4 per 50 Tk.)

```
N = 111; N/2 = 55.5 → on the ogive, across from 55.5 the curve is at about Tk 2079.
Formula: median class 2050–2100 (c.f. 38 → 68):
Median = 2050 + (55.5 − 38)/30 × 50 = 2050 + 29.17 = 2079.17 Tk
Between 2000 and 2170: 22 + 30 + 16 + (2170 − 2150)/50 × 12 = 68 + 4.8 = 72.8 ≈ 73 workers
(graphically: c.f. at 2170 ≈ 88.8 minus c.f. at 2000 = 16 → 72.8)
```

# Lesson 7 — Mode, Empirical Relation and Missing Frequencies

*pp. 19–26*

| |
|---|
| **Mode = $L + [\Delta_1/(\Delta_1 + \Delta_2)] \times h$** |
| **$\Delta_1 = f_{modal} - f_{pre\text{-}modal}$, $\Delta_2 = f_{modal} - f_{post\text{-}modal}$** |
| **Empirical relation (moderately skewed): Mode = 3 Median − 2 Mean** |

Weights 30–90 (f 18, 37, 45, 27, 15, 8): Mode = 50 + 8/(8 + 18) × 10 = 53.08. Marks (7, 32, 56, 106, 180, 164, 86, 44): Mode = 40 + 74/(74 + 16) × 10 = 48.22. Sales practice: 64.4. Ungrouped: the most frequent value (5, 8, 10, 8, 5, 8 → 8).

Empirical-relation examples (pp. 23–25): weights median 54.44, mode 53.08 → mean = (3 × 54.44 − 53.08)/2 = 55.12; marks mean 56, median 40 + (40 − 30)/22 × 20 = 49.09 → mode = 3(49.09) − 2(56) = 35.27; heights mean 145.35, median 144.92 → mode 144.06.

## Missing Frequencies (p. 26 practice 2)

With the total N and the median (and/or mode) given, write one equation from Σf = N and one from the median (or mode) formula, then solve. Practice 2 (median 33.5, mode 34, N = 230): f = 60, 100, 40.

## Exam Approach

1. Let the unknowns be $f_1$, $f_2$; Σf = N gives $f_1 + f_2 = \ldots$
2. Decide which class contains the given median; write the median equation with p.c.f. containing $f_1$ if it lies above.
3. Solve, round to whole numbers, check the median, then find the mode by the Δ formula and comment on the shape (compare mean, median, mode).

## Practice — Lesson 7

*(a) Actual — 2014-15 Q6(a) (TF p. 10). (b) Actual — 2015-16 Q6(a) (TF p. 15).*

**(a)** Incomplete distribution with total frequency 1000 and median 413.11: 300–325: 5, 325–350: 17, 350–375: 80, 375–400: ?, 400–425: 326, 425–450: ?, 450–475: 88, 475–500: 9. Estimate the missing frequencies and find the mode. (17)

**(b)** The same table with median 415. Estimate the missing frequencies, find the modal value and comment on the shape of the distribution. (18)

### ✓ Solution

**(a)** Let $f_4$ = x (375–400) and $f_6$ = y (425–450).

```
Σf = 5 + 17 + 80 + x + 326 + y + 88 + 9 = 525 + x + y = 1000  →  x + y = 475
Median 413.11 lies in 400–425; p.c.f. = 5 + 17 + 80 + x = 102 + x;  N/2 = 500
413.11 = 400 + (500 − 102 − x)/326 × 25  →  398 − x = 170.954  →  x ≈ 227,  y = 475 − 227 = 248
Mode: modal class 400–425, Δ1 = 326 − 227 = 99, Δ2 = 326 − 248 = 78
Mode = 400 + 99/(99 + 78) × 25 = 413.98
```

**(b)** Let $f_4$ = x (375–400) and $f_6$ = y (425–450).

```
Σf = 5 + 17 + 80 + x + 326 + y + 88 + 9 = 525 + x + y = 1000  →  x + y = 475
Median 415 lies in 400–425; p.c.f. = 5 + 17 + 80 + x = 102 + x;  N/2 = 500
415 = 400 + (500 − 102 − x)/326 × 25  →  398 − x = 195.6  →  x ≈ 202,  y = 475 − 202 = 273
Mode: modal class 400–425, Δ1 = 326 − 202 = 124, Δ2 = 326 − 273 = 53
Mode = 400 + 124/(124 + 53) × 25 = 417.51
```

Mean (short-cut, A = 412.5) = 413.57. Mean 413.57 < median 415 < mode 417.51: the distribution is slightly negatively skewed (longer tail to the left), though nearly symmetric.

# Quick Notes — Central Tendency

*Revision points for Lessons 5–7.*

## L5 · Means

- $\bar{x} = A + (\Sigma fd/N)h$, $d = (x - A)/h$. Marks 80 students → 56; heights → 145.35.
- G.M. = antilog($\Sigma f \log x/N$); H.M. = $N/\Sigma(f/x)$.

## L6 · Median and Quartiles

- Median = $L + (N/2 - pcf)/f \times h$; $Q_i$: $iN/4$; $D_i$: $iN/10$; $P_i$: $iN/100$.
- Ogive: across from N/2 to the curve, down to x. Wages: median 2079.17; 2000–2170: 72.8 ≈ 73.

## L7 · Mode and Missing Frequencies

- Mode = $L + \Delta 1/(\Delta 1 + \Delta 2) \times h$; Mode = 3Md − 2Mean.
- Median 413.11 → 227, 248, mode 413.98; median 415 → 202, 273, mode 417.51 (slight negative skew).